ADGIPS HACKATHON • CIVIC DISASTER TECH

Delhi Hazard Watch

Urban Flood Nowcasting, Infrastructure Risk Engine & Civic Navigation System

LIVE SYSTEM MANUAL

Project Repository: syed-i-love-you
Deployment: Render Production Cloud
Date: October 2026

Executive Summary: Delhi Hazard Watch is an AI-powered urban flood nowcasting and commuter navigation platform engineered for the National Capital Territory (NCT) of Delhi. The system combines real-time Doppler radar precipitation feeds, a 40-chronic underpass hydrodynamic model, trunk stormwater drainage graph layers, and automated safe bypass navigation routing to prevent commuter entrapment and vehicle hydro-lock during cloudburst and monsoon overflow events.

1. Live Telemetry & Weather Nowcasting Engine

🌧️ 15-Min Doppler Radar Intake

Automated integration with high-resolution meteorological models at 28.6139°N, 77.2090°E:

- **Live Precipitation Telemetry:** Real-time current downpour volume in millimeters.
- **0–3 Hour Forecast Array:** 12 discrete 15-minute Doppler telemetry buckets mapped directly into the hydraulic runoff engine.
- **Rainfall Surge Modulation:** Water accumulation across basins automatically factors live rain spikes into predictive depth calculations.

🌊 Yamuna & AQI Precursor Benchmarks

Continuous monitoring of critical regional environmental thresholds:

- **Yamuna River Gauge (Old Railway Bridge / Loha Pul):** Benchmarked against Warning Mark (204.50 m) and Danger Mark (205.33 m), referencing the historic 208.66 m record.
- **Real-Time CPCB AQI:** Continuous monitoring of particulate matter (PM2.5, PM10) to deliver commuter health warnings during heavy pollution.
- **Sensor Fallback Engine:** Redundant failover architectures ensure continuous zero-downtime availability.

2. 40-Hotspot Delhi Underpass & Basin Risk Engine

The platform models, tracks, and renders all 40 chronic flood blackspots cataloged by the Public Works Department (PWD) and Delhi Traffic Police. Water depths dynamically scale based on the active nowcast horizon:

Risk Level	Depth Threshold	Status Label	Map Pin Styling	Automated Advisory Action
LOW	< 15 cm	**Passable • Dry**	Solid Emerald (#10b981)	CLEAR: Normal passage. Drainage pumps operational. All lanes open.
MODERATE	15 cm – 40 cm	**Waterlogged • Caution**	Amber Accent (#f59e0b)	WATERLOGGED: Caution advised. Sub-surface ponding detected. Reduce vehicle speed.
CRITICAL	> 40 cm	**Inundated • Closed**	Red (#ef4444) + CSS Ping Pulse	INUNDATED: Depth exceeds 40 cm. Passage closed to vehicles. Critical hydro-lock danger.

🏛️ Central & South Corridors

- **Minto Bridge RUB:** Historic chronic chokepoint (68 cm nowcast peak).
- **Pul Prahladpur:** South-East railway underpass (78 cm peak).
- **Pragati Maidan Tunnel:** Integrated highway transit tunnel.
- **AIIMS & Moti Bagh:** Major arterial ring road underpasses.

🧭 West, North & Trans-Yamuna

- **Zakhira & Shakti Nagar:** Rohtak Road arterial underbridges.
- **Kashmere Gate ISBT:** High-density transit hub descent.
- **Loni Road & Seelampur:** Trans-Yamuna railway cuts.
- **Chilla Regulator:** Noida link road stormwater outfall zone.

Interactive Layers & Navigational Systems

3. Floating 0–3 Hour Nowcast Scrubber

Positioned at the bottom center of the viewport, the floating glass pill allows emergency managers, civil authorities, and residents to scrub forward in time across 5 predictive intervals:

Live (0h) | +30 min | +60 min | +120 min | +180 min

Clicking any interval dynamically recalculates depth curves for all 40 hotspots, toggles pulsing red ping animations for flooded passages, and syncs the Overview drawer watchlist instantaneously without page reloads.

4. Stormwater Trunk Drainage Network Graph Layer

🌊 Delhi Drainage Corridors

Rendered with dashed cyan polylines (#0284c7, weight 3.5, dashArray '6, 8'):

- **Najafgarh Drain Corridor:** 57 km main canal collecting 60% of Delhi urban discharge.
- **Barapullah Basin Drain:** Drains South Delhi (AIIMS, Lajpat Nagar, Lodhi Road).
- **Shahdara Trunk Drain:** Primary Trans-Yamuna stormwater channel.

⚠️ Outfall Surge & Backflow Warning

Automated hydrodynamic surge indicators:

- **Surge Thresholds:** Triggers visual pulsing red nodes at +60m, +120m, and +180m horizons.
- **Backflow Alerts:** Warns when high Yamuna river levels compromise gravity discharge, causing hydraulic backflow into feeder underpasses.

5. AI Flood-Safe Alternative Route Navigation

An interactive transit simulation demonstrating smart AI flood evasion between East Delhi (Vikas Marg) and Connaught Place (CP):

⛔ Default Blocked Route (Red Dotted)
Routes via ITO → Tilak Bridge → Minto Bridge. Minto Bridge is submerged under **68 cm water**. Vehicle engine stall and hydro-lock danger.

✅ Recommended Bypass (Solid Emerald)
Elevated bypass via Mandi House → Barakhamba Road corridor. **0 cm water accumulation**. Passable with only +4 min travel time.

6. Technical Architecture & Cloud Production Stack

Layer	Technology	Key Purpose & Performance Specification
Base Map Tiles	Esri World Light Gray Canvas	100% key-free, crisp light aesthetic, zero watermarks, high performance across devtunnels and mobile.
Map Engine	Leaflet.js 1.9.4	Custom HTML DivIcon pulse pins, vector polylines, bounding-box auto-frame, static mini-maps.
Frontend Shell	Vanilla ES6+ & Tailwind CSS	Single sliding drawer with multi-view state management, responsive drawer backdrop, modal alerts.
Backend API	FastAPI & Python 3.11	Async REST endpoints for zones, community reports, risk scoring, and auto-seeding SQLite database.
Cloud Hosting	Render Cloud Platform	Production Uvicorn server auto-bound to dynamic port ($PORT) with automated GitHub CI/CD deployments.